

CURRICULUM VITAE – NICHOLAS ANTON COLLINS-CRAFT

Education

DOCTOR OF PHILOSOPHY/DOCTORAT FROM THE UNIVERSITY OF SYDNEY AND ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES (2015 – 2019).

Advisors: Jointly supervised under a cotutelle agreement by Professor Itai Einav (Sydney), Professeur Jean Sulem and Dr Ioannis Stefanou (ENPC).

Thesis title: *The effect of evolving micro-structural length scale on the macroscopic constitutive behaviour of granular media*. Defended the 22nd of November, 2019.

BACHELOR OF ENGINEERING IN CIVIL ENGINEERING WITH FIRST CLASS HONOURS AND THE UNIVERSITY MEDAL FROM THE UNIVERSITY OF SYDNEY (2011 – 2014).

Honours Thesis advisor: Professor Itai Einav.

Thesis title: *The Contribution of Granular Rotation to Improving the Efficiency of Heat Transfer*.

CERTIFICATE OF ADVANCED ENGINEERING FROM THE UNIVERSITY OF SYDNEY (2011 – 2014).

Undertaken as an extension programme in parallel with the Bachelor's Degree, enabling high-achieving students to work on projects in humanitarian engineering, educational outreach and product commercialisation.

HIGHER SCHOOL CERTIFICATE AT NORTH SYDNEY BOYS HIGH SCHOOL (2005 – 2010).

Academic Employment

NOVEMBER 2024 – PRESENT Postdoctoral researcher in the geotechnical team of Laboratoire Navier at École nationale des ponts et chaussées, supervised by Professeur Jean Sulem.

SEPTEMBER 2022 – OCTOBER 2024 Marie Skłodowska-Curie Actions Postdoctoral Fellow in the Tripop team at the Inria Centre of the Université Grenoble Alpes, financed by the project LEMMA (Landslide and avalanchE Mechanics with Multiphysical data), in collaboration with Dr Vincent Acary (Inria), Dr Franck Bourrier (Inrae & Inria) and Professor Johan Gaume (WSL & ETH Zürich).

NOVEMBER 2020 – AUGUST 2022 Postdoctoral researcher in the Tripop team at the Inria Centre of the Université Grenoble Alpes, financed by the Region Auvergne-Rhône-Alpes project SMART-PROTECT, supervised by Dr Vincent Acary and Dr Franck Bourrier, in collaboration with industrial partner Géolithe.

JUNE 2015 – NOVEMBER 2019 Doctoral student at the University of Sydney and École nationale des ponts et chaussées, financed by an Australian Postgraduate Award and a University of Sydney Merit top-up scholarship.

Journal Articles

[ON THE MOREAU–JEAN SCHEME WITH THE FRÉMOND IMPACT LAW: ENERGY CONSERVATION AND DISSIPATION PROPERTIES FOR ELASTODYNAMICS WITH CONTACT, IMPACT AND FRICTION](#)

V. Acary & N.A. Collins-Craft, in the Journal of Theoretical, Computational and Applied Mechanics, June 2025.

[ON THE FORMULATION AND IMPLEMENTATION OF EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT](#)

N.A. Collins-Craft, F. Bourrier & V. Acary, in Computer Methods in Applied

Mechanics and Engineering, Volume 400, October 2022.

[A COSSERAT BREAKAGE MECHANICS MODEL FOR BRITTLE GRANULAR MEDIA](#)

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, in Journal of the Mechanics and Physics of Solids, Volume 141, August 2020.

Preprints

[ON THE FORMULATION AND IMPLEMENTATION OF MIXED MODE I AND MODE II EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT AND FRICTION](#)

N.A. Collins-Craft, V. Acary. Under review at Computer Methods in Applied Mechanics and Engineering.

[THE INFLUENCE OF GRAIN CRUSHING AND PORE COLLAPSE ON THE FORMATION OF FAULTS](#)

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav. Under review at Journal of Geophysical Research: Solid Earth.

**Working Papers
(available on
demand)**

[MONOLITHIC MECHANICAL MODEL INCORPORATING SOLID CONTACT FOR ANISOTROPIC POROUS MATERIALS](#)

L. Zhang, N.A. Collins-Craft & P. Braun.

**Conference
Presentations
(presenter
underlined)**

[PREDICTING THE MICROSTRUCTURAL EVOLUTION OF SHEAR BANDS IN CRUSHABLE GRANULAR MEDIA](#)

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, 8th International Conference on Coupled THMC Processes: Geomechanics for Energy and Environmental Applications, Nicosia, Cyprus, 20/10/2025 – 23/10/2025.

[MICROMECHANICAL POROUS MEDIA SIMULATION INCORPORATING SOLID CONTACT THROUGH ENHANCED LINEAR COMPLEMENTARITY PROBLEM METHODOLOGY](#)

L. Zhang, N.A. Collins-Craft, J.-M. Pereira & P. Braun, 3rd International Conference on Energy Geotechnics, Paris, France, 17/06/2025 – 19/06/2025.

[A NEW MOREAU–JEAN SCHEME THAT GUARANTEES DISSIPATIVITY](#)

N.A. Collins-Craft, V. Acary & N.A. Collins-Craft, 11th Contact Mechanics International Symposium, Lyon, France 22/05/2024 – 24/05/2024.

[EXTENDING NON-SMOOTH CONTACT MECHANICS TO COHESIVE ZONE MODELLING](#)

N.A. Collins-Craft, F. Bourrier & V. Acary, 7th International Conference on Computational Contact Mechanics, Turin, Italy, 05/07/2023 – 07/07/2023.

[A NON-SMOOTH EXTRINSIC COHESIVE ZONE MODEL INCLUDING CONTACT AND FRICTION](#)

N.A. Collins-Craft, F. Bourrier & V. Acary, The Seventh International Conference on Computational Modelling of Fracture and Failure of Materials and Structures, Prague, Czechia, 21/06/2023 – 23/06/2023.

[A NON-SMOOTH COHESIVE ZONE MODEL FOR ROCK FRACTURE AND CONTACT](#)

N.A. Collins-Craft, F. Bourrier, J. Gaume & V. Acary, European Geophysical Union General Assembly 2023, Vienna, Austria, 23/04/2023 – 28/04/2023.

[A NON-SMOOTH FORMULATION OF AN EXTRINSIC COHESIVE ZONE MODEL WITH CONTACT AND FRICTION](#)

N.A. Collins-Craft, F. Bourrier & V. Acary, 11th European Solid Mechanics Con-

ference, Galway, Ireland, 04/07/2022 – 08/07/2022.

AN EXTRINSIC COHESIVE ZONE MODEL WITH CONTACT AND FRICTION

N.A. Collins-Craft, F. Bourrier & V. Acary, 18th European Mechanics of Materials Conference, Oxford, United Kingdom, 04/04/2022 – 06/04/2022.

AN EXTRINSIC COHESIVE ZONE MODEL WITH CONTACT DEVELOPED IN THE NON-SMOOTH MECHANICS FRAMEWORK

N.A. Collins-Craft, F. Bourrier & V. Acary, 33rd Nordic Seminar on Computational Mechanics, Jönköping, Sweden, 25/11/2021 – 26/11/2021.

UNIFYING BREAKAGE MECHANICS WITH THE COSSERAT CONTINUUM TO PREDICT SHEAR BAND LOCALISATION

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, International Congress of Theoretical and Applied Mechanics 2020+1, Milan, Italy, 22/08/2021 – 27/08/2021.

A THEORY TO PREDICT SHEAR BAND FORMATION IN GRANULAR MEDIA WITH EVOLVING GRAIN SIZE DISTRIBUTIONS

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Micro2Macro 2018, Reggio Calabria, Italy, 29/05/2018 – 01/06/2018.

A FORMULATION OF BREAKAGE MECHANICS IN THE COSSERAT CONTINUUM TO PREDICT SHEAR BAND FORMATION IN GRANULAR MEDIA

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, 12th EURO-conference on Rock Physics & Geomechanics, Ma'ale HaHamisha, Israel, 05/11/2017 – 10/11/2017.

**Accepted
Conference
Presentations
(presenter
underlined)**

ACCOUNTING FOR GRAIN CRUSHING AND PORE COLLAPSE FOR STRAIN LOCALIZATION IN ROCKS

N.A. Collins-Craft, J. Sulem, I. Stefanou & I. Einav, 15th International Symposium on Continuum Models and Discrete Systems, Rome, Italy 08/06/2026 – 12/06/2026.

**Conference
Posters**

THE EFFECT OF GRAIN SIZE REDUCTION ON SHEAR BAND FORMATION

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Patterns in Geomechanics, Sydney, Australia, 29/01/2019 – 01/02/2019.

THE EFFECT OF GRAIN SIZE REDUCTION DURING SHEARING OF GRANULATED FAULT GOUGE

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, AGU Fall meeting, Washington D.C., United States of America, 10/12/2018 – 14/12/2018.

MODELLING CRUSHABLE GRANULAR MEDIA IN THE COSSERAT CONTINUUM

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, Navier Seminar, Paris, France, 05/03/2018.

PROGRESS TOWARDS A CONSTITUTIVE MODEL FOR CRUSHABLE GRANULAR MEDIA USING COSSERAT CONTINUUM MODELLING

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav, ERC REALISM Kick-off Meeting, Paris, France, 15/03/2017.

Invited Seminars

NONSMOOTH MECHANICS FOR FRACTURE AND FRICTION PROBLEMS IN GEOMATERIALS

CivMin seminar, University of Toronto, 14/07/2025.

NONSMOOTH MECHANICS FOR FRACTURE AND FRICTION PROBLEMS IN GEOMATERIALS
SPREE seminar, Northwestern University, 28/02/2024.

LA PERSPECTIVE MOREAU-ISTE SUR LA FRACTURE : EXTENSION DE LA MÉCANIQUE NONSMOOTH AUX MODÈLES DE ZONES COHÉSIVES EXTRINSÈQUES AVEC CONTACT ET FRICTION
laboratory seminar, Université de Montpellier, 18/01/2024.

NON-SMOOTH MECHANICS FOR FRACTURE PROPAGATION IN GEOMATERIALS
ENAC Seminar Series, École Polytechnique Fédérale de Lausanne, 16/05/2023.

LA MÉCANIQUE NON-LISSE POUR LA MODÉLISATION DE LA FISSURATION : PRINCIPES, PREMIERS RÉSULTATS ET PERSPECTIVES
MISES seminar, Sorbonne Université, 08/03/2023.

FORMULATING AND IMPLEMENTING EXTRINSIC COHESIVE ZONE MODELS IN THE NON-SMOOTH MECHANICS FRAMEWORK
CERMES seminar, École nationale des ponts et chaussées, 22/10/2022.

PREDICTING STRAIN LOCALISATION IN CRUSHABLE GRANULAR MEDIA USING THE COSSERAT CONTINUUM
TRIPOP seminar, Inria Grenoble-Rhône-Alpes, 07/12/2020.

A COSSERAT BREAKAGE MECHANICS MODEL FOR BRITTLE GRANULAR MEDIA and AN INTRODUCTION TO JULIA
CoQuake seminar, École centrale de Nantes, 02/06/2020.

THE EFFECT OF EVOLVING MICRO-STRUCTURAL LENGTH SCALE ON THE MACROSCOPIC CONSTITUTIVE BEHAVIOUR OF GRANULAR MEDIA
Granular Forum, the University of Sydney, 20/06/2018.

THE EFFECT OF EVOLVING MICRO-STRUCTURAL LENGTH SCALE ON THE MACROSCOPIC CONSTITUTIVE BEHAVIOUR OF GRANULAR MEDIA
CERMES seminar, École nationale des ponts et chaussées, 19/02/2018.

THE EFFECT OF EVOLVING MICRO-STRUCTURAL LENGTH SCALE ON THE MACROSCOPIC CONSTITUTIVE BEHAVIOUR OF GRANULAR MEDIA
ENS seminar, École normale supérieure de Paris, 03/03/2017.

**Seminars to
Masters Students**

L'EFFET DE L'ÉVOLUTION DE L'ÉCHELLE DE LONGUEUR MICRO-STRUCTUREL SUR LE COMPORTEMENT MACROSCOPIQUE CONSTITUTIF DES MILIEUX GRANULAIRES
MSROE seminar, École nationale des ponts et chaussées, 11/01/2018. Approximately 30 students.

THE EFFECT OF EVOLVING MICRO-STRUCTURAL LENGTH SCALE ON THE MACROSCOPIC CONSTITUTIVE BEHAVIOUR OF GRANULAR MEDIA
MSROE seminar, École nationale des ponts et chaussées, 05/01/2017. Approximately 30 students.

**Seminars to
Undergraduate
Students**

MODÉLISATION DU CONTACT, DE L'IMPACT, DU FROTTEMENT ET DE LA FRACTURE

“Métiers de la recherche” seminar, École nationale des ponts et chaussées, 24/01/2025.
Two groups of ten students each.

**Standalone
Software**

[JULIA CONSTITUTIVE MODEL INTEGRATION](#)

Codes that use the Julia language to integrate the model described in “A Cosserat Breakage Mechanics Model for Brittle Granular Media”, as well as conduct linear stability analyses over given loading paths. Examples of model outputs and linear stability analyses are saved in a Zenodo deposit.

**Contributions to
Standalone
Software
Primarily
Authored by
Others**

[NUMERICAL GEOLAB](#)

Software primarily authored by Ioannis Stefanou and Alexandros Stathas. Codes that provide a layer over the FEniCS finite element software to facilitate its usage for nonlinear, multiphysical and higher-order continuum mechanics problems. I am currently the active maintainer of the repository as well as developing new material models and new features such as the ability to include models written in the Julia programming language.

**Software
Corresponding to
a Paper**

[THE INFLUENCE OF GRAIN CRUSHING AND PORE COLLAPSE ON THE FORMATION OF FAULTS](#)

Codes accompanying the preprint of the same name that install the same computational environment as was used in the paper, and that are sufficient to reproduce all simulations and figures.

[ON THE FORMULATION AND IMPLEMENTATION OF MIXED MODE I AND MODE II EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT AND FRICTION](#)

Codes accompanying the preprint of the same name that are sufficient to install the necessary software (with the exact versions we used) and reproduce all simulations and plots thereof in the paper.

[ON THE MOREAU–JEAN SCHEME WITH THE FRÉMOND IMPACT LAW: ENERGY CONSERVATION AND DISSIPATION PROPERTIES FOR ELASTODYNAMICS WITH CONTACT, IMPACT AND FRICTION](#)

Codes accompanying the article of the same name that are sufficient to reproduce all the simulations bar two, which are available in the siconos-tutorials repository. These codes are also able to reproduce all figures in the paper, provided two images that are stored in the accompanying data deposit are present.

[ON THE FORMULATION AND IMPLEMENTATION OF EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT](#)

Codes accompanying the article of the same name that are sufficient to reproduce all simulations and figures in the paper.

Data

[THE INFLUENCE OF GRAIN CRUSHING AND PORE COLLAPSE ON THE FORMATION OF FAULTS](#)

The results of all simulations conducted in the preprint of the same name, both single material point and full finite element simulations, as well as calculations of the yield surface of the new and original model under various values of material parameters and state variables.

[ON THE FORMULATION AND IMPLEMENTATION OF MIXED MODE I AND MODE II EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT AND FRICTION](#)

The results of all simulations conducted in the preprint of the same name.

ON THE MOREAU–JEAN SCHEME WITH THE FRÉMOND IMPACT LAW: ENERGY CONSERVATION AND DISSIPATION PROPERTIES FOR ELASTODYNAMICS WITH CONTACT, IMPACT AND FRICTION

The results of all simulations conducted in the article of the same name.

JULIA CONSTITUTIVE MODEL INTEGRATION REPLICATION FILES

The results of four simulations and corresponding localisation analysis codes that are included in the Julia constitutive model integration repository.

ON THE FORMULATION AND IMPLEMENTATION OF EXTRINSIC COHESIVE ZONE MODELS WITH CONTACT

The results of all simulations conducted in the article of the same name, as well as the finite element meshes used.

Students supervised

2024 – CHLOÉ GERGELY, masters student in the program *Master Mathématiques et applications* of the *Université Claude Bernard Lyon 1*, co-supervised with Franck Bourrier & Vincent Acary. Continued on to a PhD in the group.

Working Papers

A NOVEL MULTISCALE METHOD AND ITS APPLICATION TO FAULTS

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav.

SHEAR BAND FORMATION IN THE SAINT-MAXIMIN LIMESTONE MODELLED WITH BREAKAGE MECHANICS

N.A. Collins-Craft, C. Doré-Ossipyan & J. Sulem.

THERMO-HYDRO-MECHANICAL COSSERAT BREAKAGE MECHANICS MODEL

N.A. Collins-Craft, I. Stefanou, J. Sulem & I. Einav.

MONOLITHIC HYDROMECHANICAL MODEL FOR FLUID FLOW AND SOLID DEFORMATION

L. Zhang, N.A. Collins-Craft & P. Braun.

COMPARISON OF THE PREDICTIONS OF A FRACTURE MODEL WITH EXPERIMENTAL RESULTS OBTAINED ON LIMESTONE SAMPLES

N.A. Collins-Craft & C. Silvani.

Prizes, Scholarships and other Academic Achievements

2022 Marie Skłodowska-Curie Actions Postdoctoral Award.

2017 Early Career Researcher grant to attend the 12th Euro-conference on Rock Physics & Geomechanics.

2016 Postgraduate Research Support Scheme.

2015 Australian Postgraduate Award, University of Sydney Merit Award (for three and a half years each).

2014 D.G. Walkom Prize for First Class Honours in Civil Engineering, Dean's List of Excellence in Academic Performance.

2013 D. Campbell-Allen Prize in Civil Engineering, invitation to participate in the Summer Research Scholarship programme.

2012 University of Sydney Academic Merit Prize, Dean's List of Excellence in Academic Performance, invitation to participate in the Summer Research Scholarship programme.

- 2011 University of Sydney Academic Merit Prize, Dean's List of Excellence in Academic Performance, invitation to participate in the Summer Research Scholarship programme.
- 2010 Awarded Roads and Maritime Services Scholarship in Civil Engineering for length of Bachelor's Degree.

Academic Service REVIEWER for Acta Geotechnica, Computer Methods in Applied Mechanics and Engineering, Computational Mechanics, Computers and Geotechnics, International Journal of Solids and Structures, Interpore Journal, Journal of Geophysical Research: Solid Earth, Mathematics in Engineering, Powder Technology, Rock Mechanics and Rock Engineering, and Scientific Reports.

ORGANISATIONAL ASSISTANT for 6th International Conference on Coupled THMC Processes in Geosystems, Paris, France, 05/07/2017 – 07/07/2017 and 3rd International Conference on Energy Geotechnics, Paris, France, 17/06/2025 – 19/06/2025.

Other Academic Qualifications QUALIFICATION AUX FONCTIONS DE MAÎTRE DE CONFÉRENCES in sections 36 and 60 of the Conseil National des Universités, 2022 qualification campaign.

Other Academic Work 2025 Tutor for *Programme Étudiants Réfugiés* (Refugee Students Program), one-to-one mathematics and informatics tutoring of refugee students to help bring them to the required level for further study in France.

2023 Tutor for MAP101 *Analyse élémentaire et introduction au calcul scientifique* (Elementary real analysis and introduction to scientific computing), first year applied mathematics and informatics course at Université Grenoble Alpes.

2019 Assignment marking and substitute tutor for CIVL2410 Soil Mechanics, second year civil engineering course at the University of Sydney.

2017 – 2018 Scientific English editing of a wide variety of scientific papers written by other ENPC researchers.

2016 Tutor for ENGG1801 Engineering Computing, first year general engineering course at the University of Sydney.

2015 Tutor for CIVL3411 Geotechnical Engineering, third year civil engineering course at the University of Sydney.

Non-Academic Employment ENGLISH TUTOR, 2017 – 2018. Self-employed English tutor in France.

KAMBALA, 2015 – 2016. Debating coach for the Independent Schools Debating Association and Archdale Schools competitions.

THE ARTS UNIT, 2011 – 2015. Debating adjudicator for public primary and high schools in the Northern Sydney region.

MATHS AND PHYSICS TUTOR, 2015. Self-employed private tutor for first year Maths and Physics subjects.

ROADS AND MARITIME SERVICES, 2011 – 12, 2012 – 13, 2013 – 14. Three separate stints over the summer break in association with my scholarship. I played a role in project management of a A\$400 million public works (Northern Beaches Hospital Connectivity and Network Enhancement) in the Project Development unit (13 – 14), developer liaison on large projects such as Wynyard Walk and Moorebank

Units Relocation (Holsworthy Barracks) in the Project Management unit (12 – 13), and material behaviour research in the Pavements unit (11 – 12).

TROMBONE TUTOR, 2011 – 14. Self-employed.

ENGINEERING STUDIES TUTOR, 2011. Self-employed.

Technical Skills

THEORETICAL AND NUMERICAL CONSTITUTIVE MODELLING

Specialisation in modelling the localisation of deformation. Higher order continuum mechanics, in particular the Cosserat continuum, linear stability analysis of mechanical systems, cohesive zone modelling of crack propagation.

PROGRAMMING

Proficient in Python, Julia, MATLAB and Mathematica, competent in C++, some notions in Haskell, Fortran and Basic.

SOFTWARE

Proficient in FEniCS, Siconos, $\text{\LaTeX}$ and Microsoft Office suite, competent in GIMP, some notions in MOOSE.

OPERATING SYSTEMS

Proficient in Microsoft Windows, Ubuntu Linux.

Citizenship

Australian.

Languages

English (mother tongue), French (proficient).

Contact Information

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